

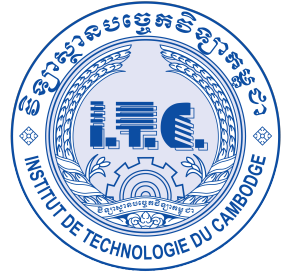


Industrial Organization

Lecture 4

- Capacity Management -

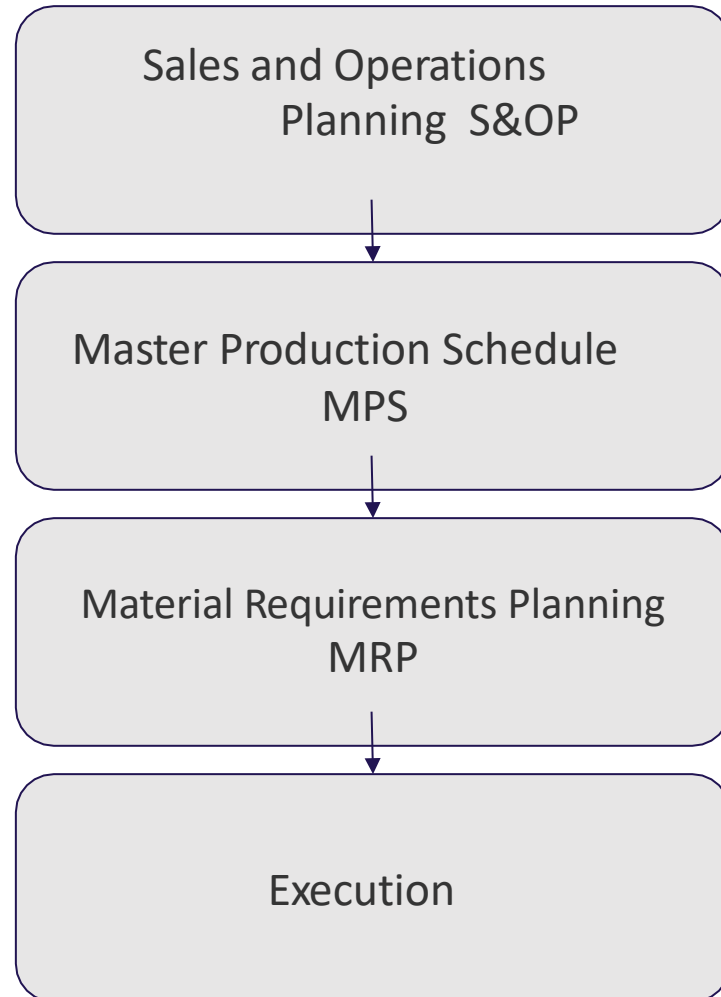
Course objectives



- Understand the link between **Capacity & Priority planning**
- Describe the process of **Capacity Planning**
- Explain why **standard hours** are used to **measure capacity**
- Explain the five elements of **manufacturing lead time**
- Calculate **available capacity** and **load**
- Compare capacity v/s load and resolve **issues**

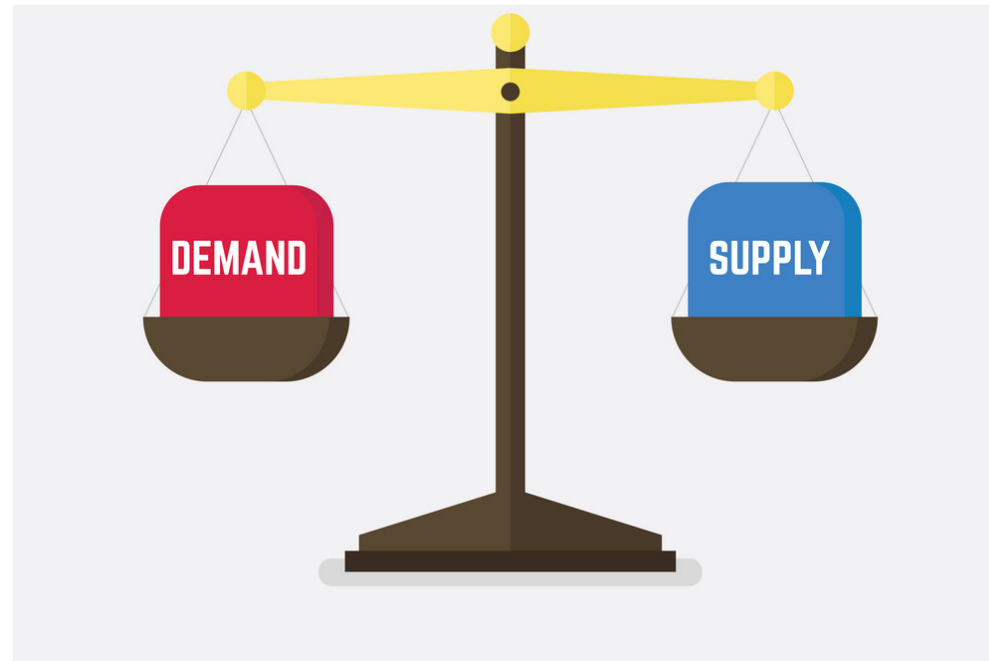
1. Reminders from Industrial Planning
2. Capacity Management
3. Capacity Requirements Planning
4. Example of calculations

1/ Reminders from Industrial Planning

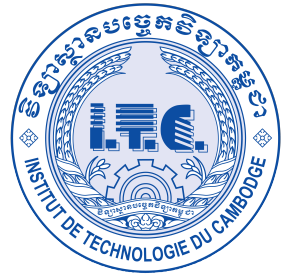


- **Mid- to long-term** planning at **product family** level
- Translated **business** plan into **production** plan
- **Short- to mid-term** planning
- Detailed planning at the **end-item level**
- **Short-term** planning
- Detailed planning at the **end-item or product-component** level
- Planned and confirmed **orders**; stock levels are known
- **Execution** horizon (hours, days)
- **Current** vision of production execution

ULTIMATE OBJECTIVE OF INDUSTRIAL AND CAPACITY PLANNING:



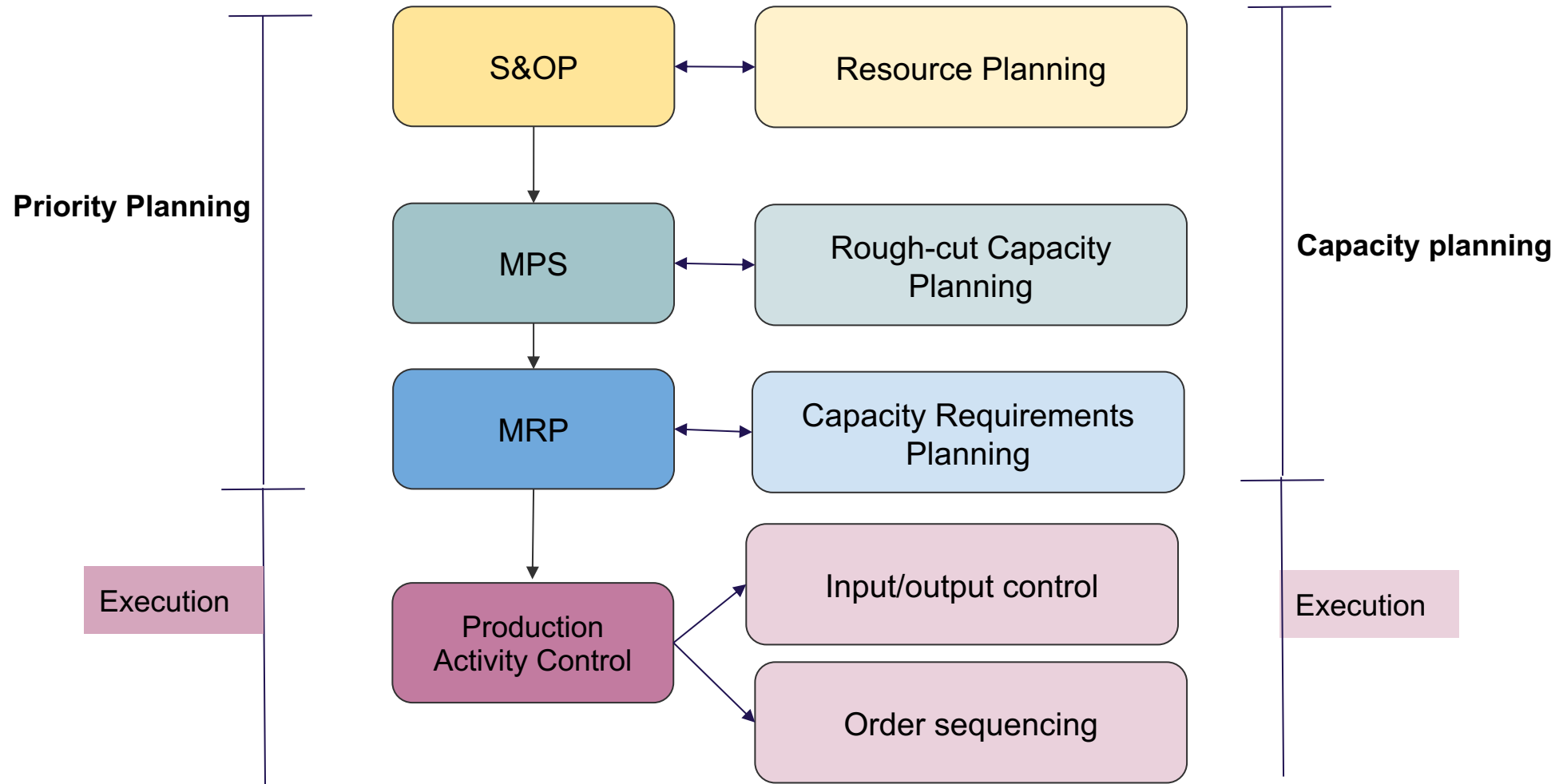
Priority Planning... must be validated with Capacity Planning



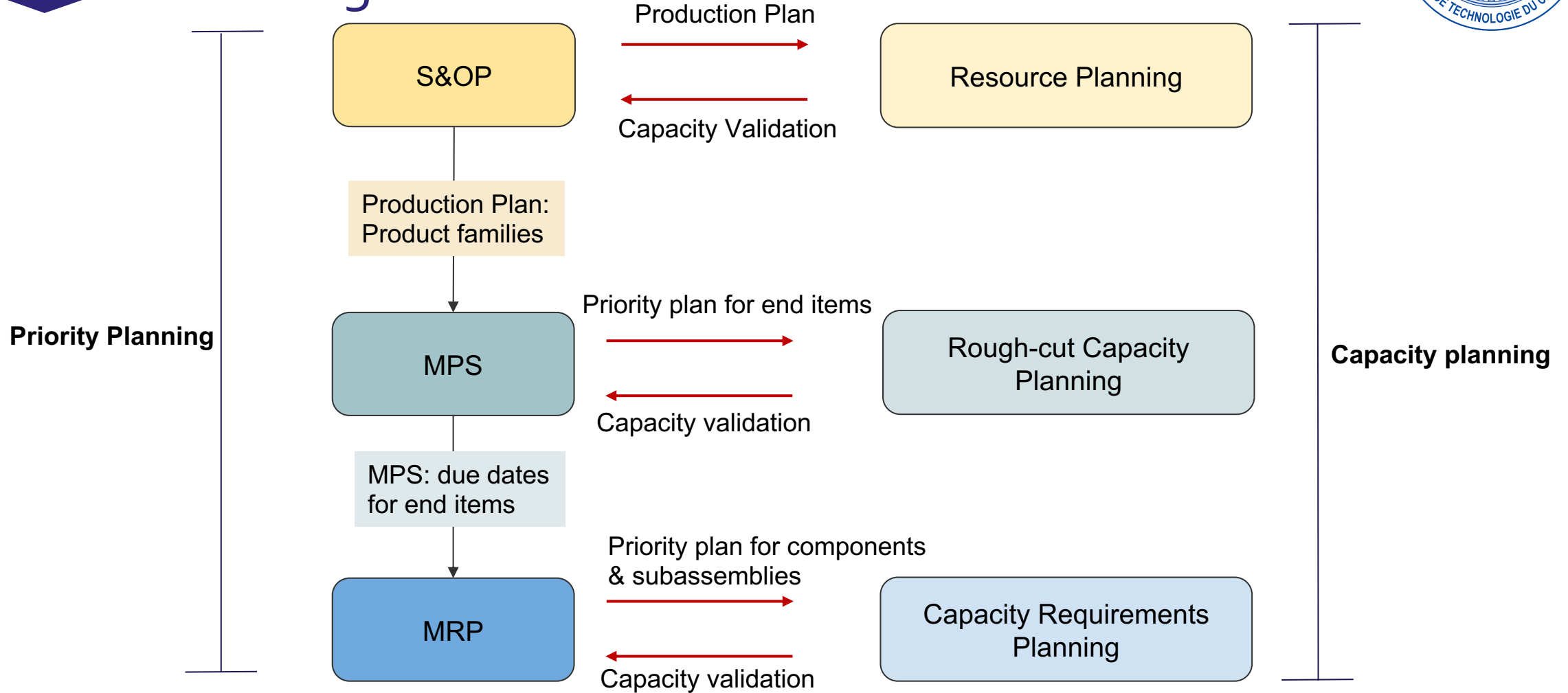
2 mandatory rules:

- At **each level** of the industrial Planning → make sure that there are **enough resources** to achieve the **production plan**
- Always check how much **capacity is available** and how much is **required (load)**, before performing a planning of the resources based on the production schedule

Priority Planning... must be validated to Capacity Planning



Priority Planning... must be validated with Capacity Planning

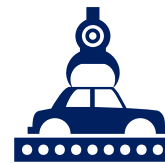


The capacity that is **available** and the capacity that is **required** must be determined at different levels:

■ Resource level: Machine or worker



■ Work center level: a production line



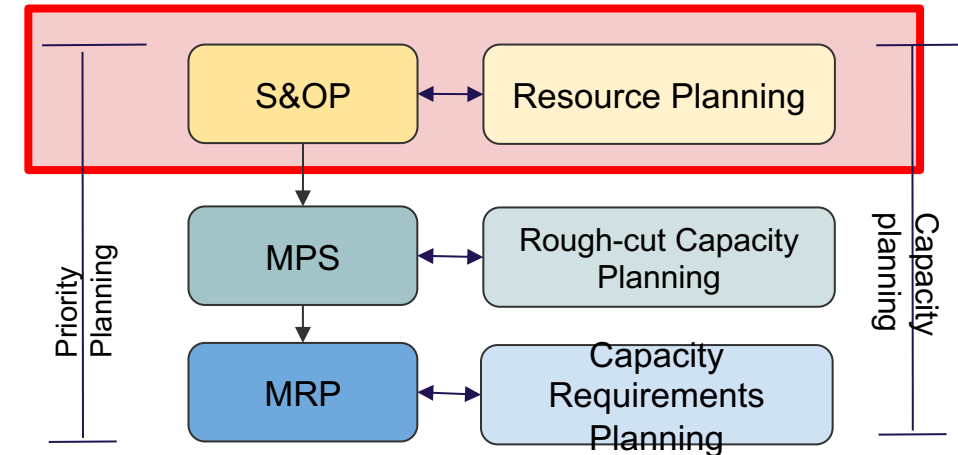
■ Plant level: several work centers



Different levels of Capacity Planning, linked to levels of Priority Planning

RESOURCE PLANNING (RP):

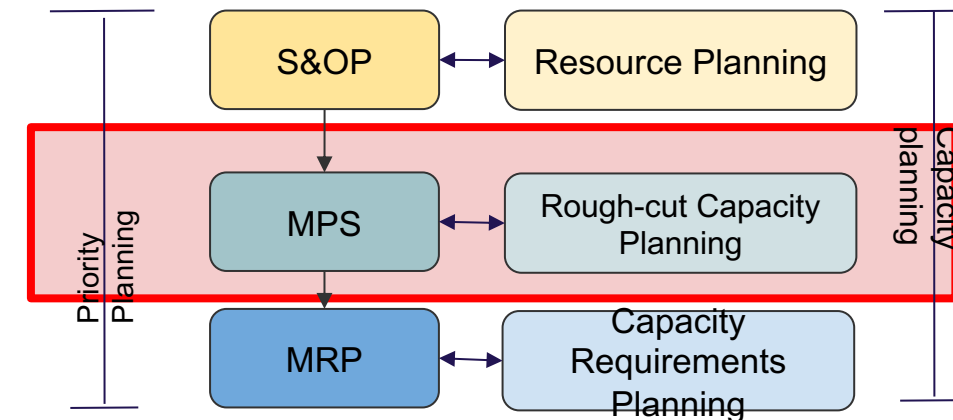
- Linked to **S&OP**
- At **Product Family** level
- Mid- to long-term **capacity requirements**
- Aims at identifying **critical resources**
- Can involve mid- to long-term **investment**
- If Resource Planning matches → GO MPS
- Else: review plans



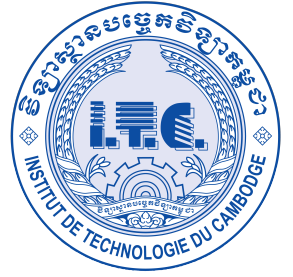
Different levels of Capacity Planning, linked to levels of Priority Planning

ROUGH-CUT CAPACITY PLANNING (RCCP)

- Linked to **MPS**
- At **end-item** product level
- Short- to mid-term **capacity requirements**
- Checks the **balance** between load & capacity
- Focuses on **bottlenecks & critical resources**
- Can involve mid- to long-term **investment**
- Uses the **routing**

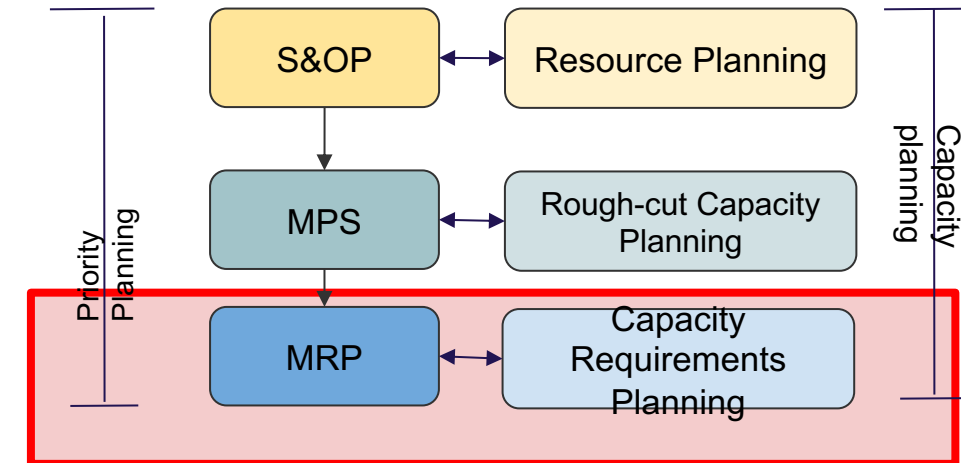


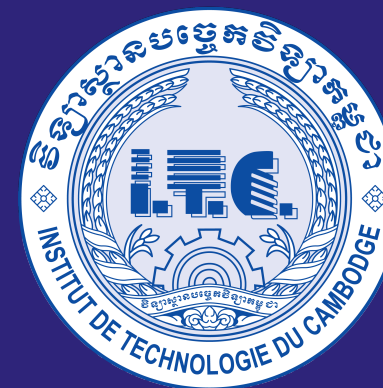
Different levels of Capacity Planning, linked to levels of Priority Planning



CAPACITY REQUIREMENTS PLANNING (CRP)

➔ *More detail later*





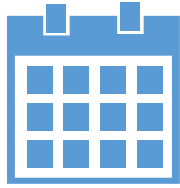
2/ Capacity Management

■ Definition:

- “The function of **establishing, measuring, monitoring and adjusting limits or levels of capacity** in order to **execute** all manufacturing schedules.
- Capacity management is executed at **4 levels**: RP (*Resource Planning*), RCCP (*Rough-Cut-Capacity Planning*), CRP (*Capacity Requirements Planning*), and input/output control”

[APICS Dictionary](#)

Capacity Management consists of 2 phases



PLANNING PHASE

- **Determine resources** needed for the priority plan
- Validate they will be **available**
- This phase occurs at **S&OP, MPS and MRP**



CONTROLLING PHASE

- **Monitor output** and **compare** it to plan
- Make **adjustments** if necessary
- This phase occurs during **Production Activity Control** (Execution)

■ Definition:

- “The **capability** of a system to **perform its expected function**
- The **capability** of a worker, machine, work center, plant, or organization to **produce output per time period.**”

[APICS Dictionary](#)

Capacity Planning Process



DETERMINE

- Determine the **capacity available**



TRANSLATE

- Translate the **priority plan into capacity required** for each time period



SUM & COMPARE

- Sum up capacities for each time **period required for each resource** in order to **compare with capacity available**



RESOLVE

- Resolve **differences** between **available capacity and required capacity** for each time period

But... what is capacity available?

■ Definition

- “The **capability** of a system or resource to produce a **quantity of output** in a particular time period”

[APICS Dictionary](#)

■ 2 types of “Capacity available”:

RATED CAPACITY: calculated

Calculated using the following information:

- Available time
- Utilization
- Efficiency

DEMONSTRATED CAPACITY: history

Demonstrated from historical data

Determining Capacity available with the rated capacity

- **Rated capacity** is the expected output capability of a resource or system, based on available time, efficiency, utilization

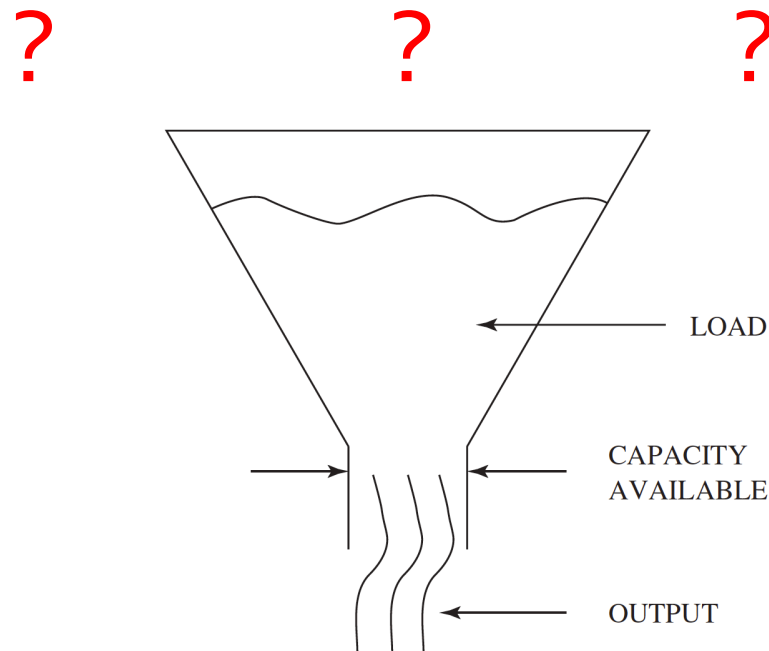


FIGURE 5.1 Capacity versus load.

Determining Capacity available with the rated capacity

AVAILABLE TIME: number of hours a work center **can be used**.

*(note: absenteeism, trainings, meetings, breaks, etc. must be taken into account, meaning they must be **included** in the available time).*

Composed of 2 major elements:

- Number of machines or operators
- Hours of operations (hours per day, days per week, weeks per year)

AVAILABLE TIME

= number of machines (or workers) × hours per day × days per week

➔ What is the weekly available time for a work center that has 4 machines, works 8h/d for 5d/w?

Determining Capacity available with the rated capacity

UTILIZATION: the % of time a work center is active compared to the available time

*(note: absenteeism, trainings, meetings, breaks, etc. must be **subtracted** from the available time).*

Composed of 2 elements:

- Hours **actually** worked
- Hours of available time

$$Utilization (\%) = \frac{Hours\ actually\ worked}{Hours\ of\ available\ time} \times 100$$

➔ What is the utilization rate of a work center whose available time is of 160h/w but whose actual hours worked is 120h?

Determining Capacity available with the rated capacity

EFFICIENCY: the output as compared to the standard. The workers might be working at a faster or slower pace than the standard working pace → efficiency can be more or less than 100%

Composed of 2 elements:

- Standard hours of output
- Hours actually worked

$$\text{Efficiency (\%)} = \frac{\text{Standard hours of work produced}}{\text{Hours actually worked}} \times 100$$

→ What is the efficiency rate of a work center where standard hours produced for a week is 150h, but actual hours is 120?

Determining Capacity available with the rated capacity

- Rated capacity is the expected output capability of a resource or system, based on available time, efficiency, utilization:

$$\begin{aligned} &\textit{Rated capacity}(\textit{standard hours}) \\ &= \textit{available time (h)} \times \textit{efficiency (\%)} \times \textit{utilization (\%)} \end{aligned}$$

- ➔ What is the rated capacity of 3 machines that are operated 8h/d for 5d/w, knowing that Utilization has been 75% and efficiency has been 125%?
- ➔ What happens if available capacity is overstated?

But... what is capacity available with regards to demonstrated capacity?

■ Definition

- “The **capability** of a system or resource to produce a **quantity of output** in a particular time period”

[APICS Dictionary](#)

■ 2 types of “Capacity available”:

RATED CAPACITY: calculated

Calculated using the following information:

- Available time
- Utilization
- Efficiency



DEMONSTRATED CAPACITY: history

Demonstrated from historical data

Determining Capacity available with the demonstrated capacity

Demonstrated capacity: examines the previous production records and uses that information as the available capacity.

Composed of 2 elements:

- Historical records of resource output capability
- Average of the resource's past period output

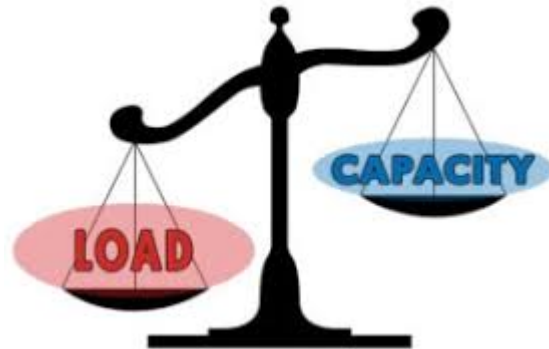
$$\text{Demonstrated capacity}(\text{standard hours of work}) \text{ over } n \text{ past period} \\ = \frac{\sum_{i=1}^n (\text{Standard hours})}{n}$$

→ What is the demonstrated capacity of a work center where it produced 120, 130, 150 and 140 standard hours per week for the previous 4 weeks?

Determining Capacity available (rated or demonstrated)... so what?

→ Compare it to the capacity **required**: the load

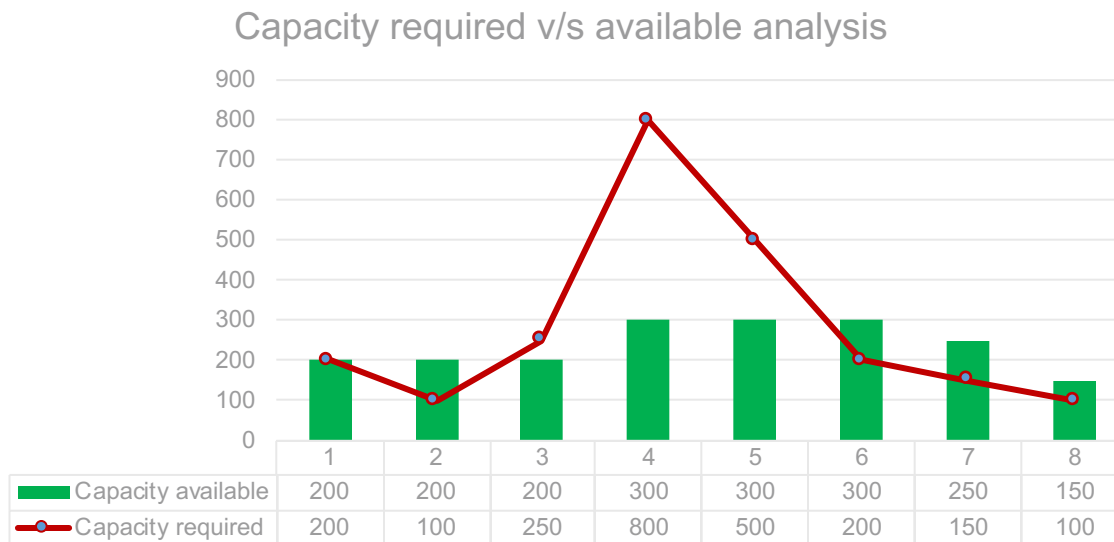
- Load = Capacity required: the amount of **planned work** scheduled for a resource or several resources (usually expressed in hours or nb of pieces)



Determining Capacity available (rated or demonstrated)... so what?

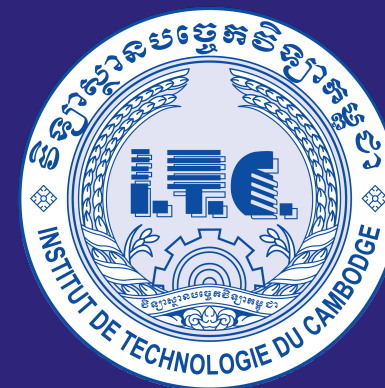
→ Compare it to the capacity **required**: the load

- Example:



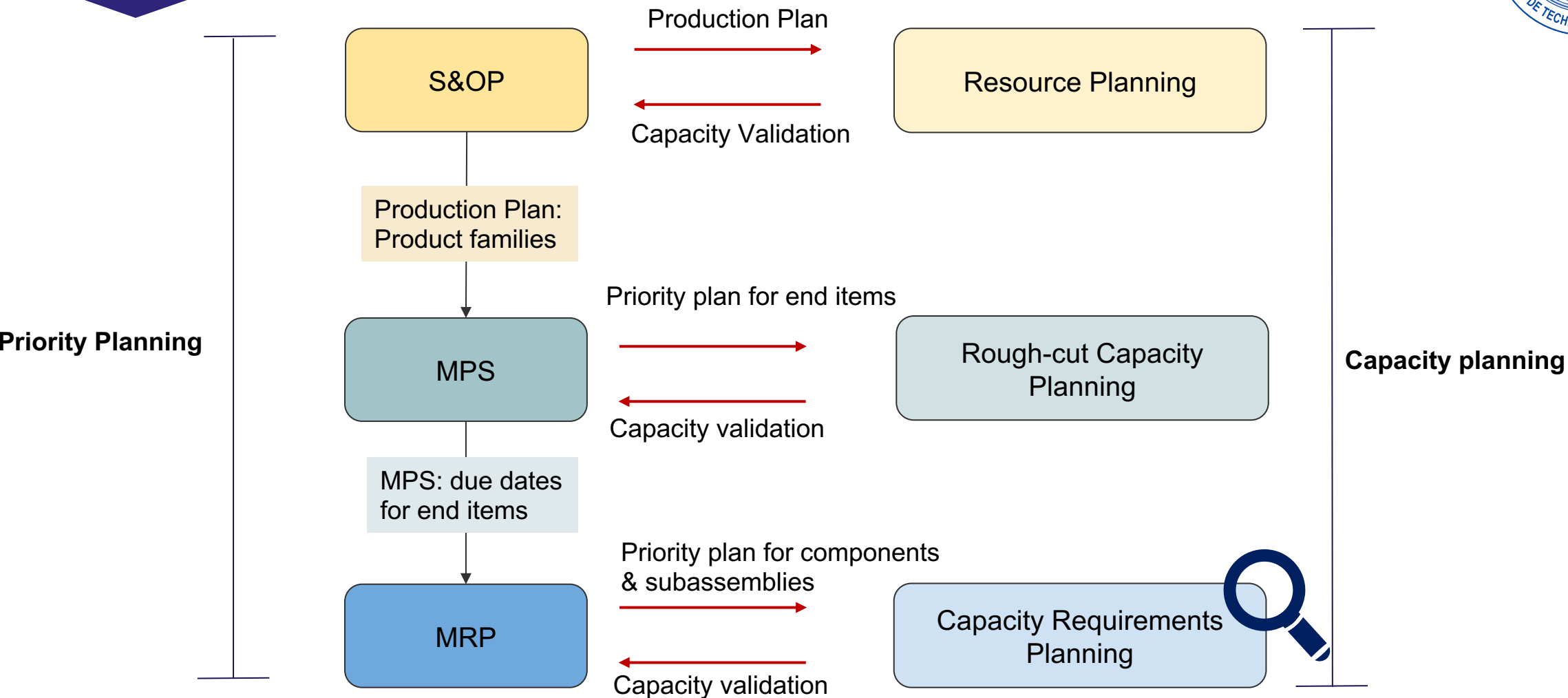
Conclusion of the analysis:
production line is **overloaded**

Capacity management mostly aims at finding the solution:
HOW TO ABSORB THE EXTRA LOAD IN ORDER TO MEET THE DEMAND?

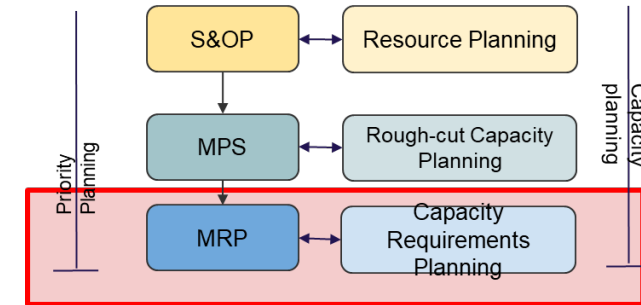


3/ Capacity Requirements Planning

Capacity Planning Hierarchy



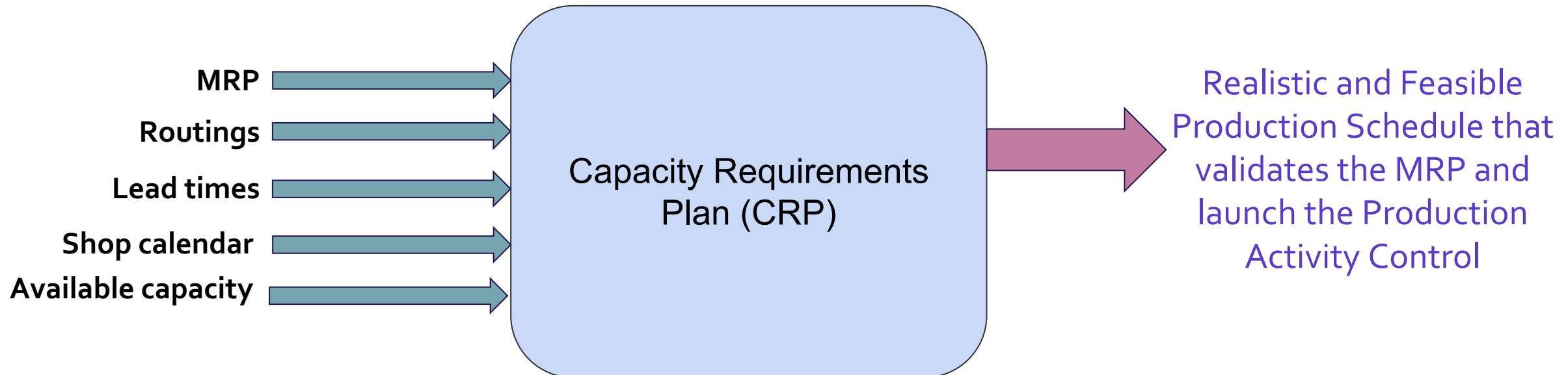
Key inputs to CRP (Capacity Requirements Planning)



INPUTS

Capacity planning process at MRP level

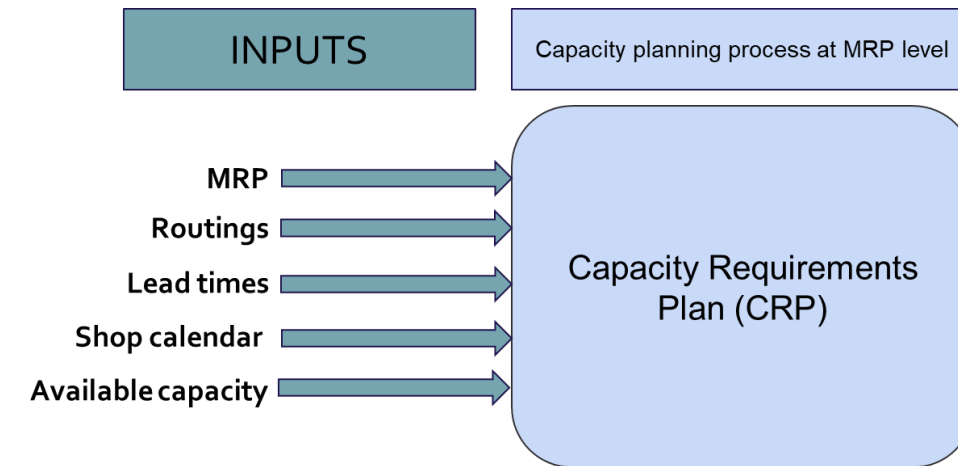
OUTPUTS



Key inputs to CRP (Capacity Requirements Planning)

Definition / usage of the inputs:

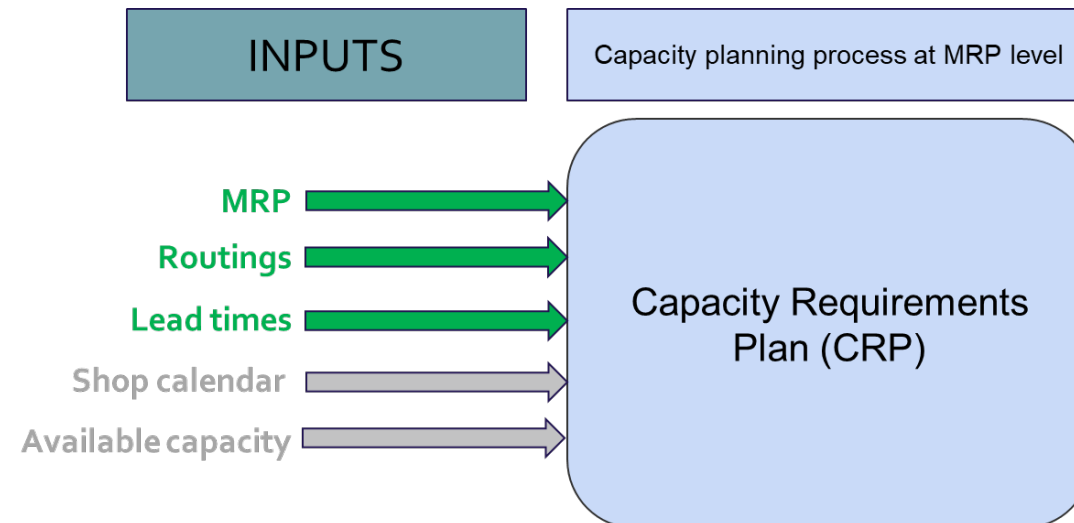
- **MRP** : Planned order release for each item
- **Routing file**: time needed & where work is done for each manufactured item
- **Lead times**: production lead time (*see later*)
- **Shop calendar**: number of working day available
- **Available capacity**: cf. previous section



Key input to CRP: the load

■ How to transform this into
'Load'?

➔ Combination of MRP,
Routings & Lead times



Key input to CRP: the load → how to calculate the load?

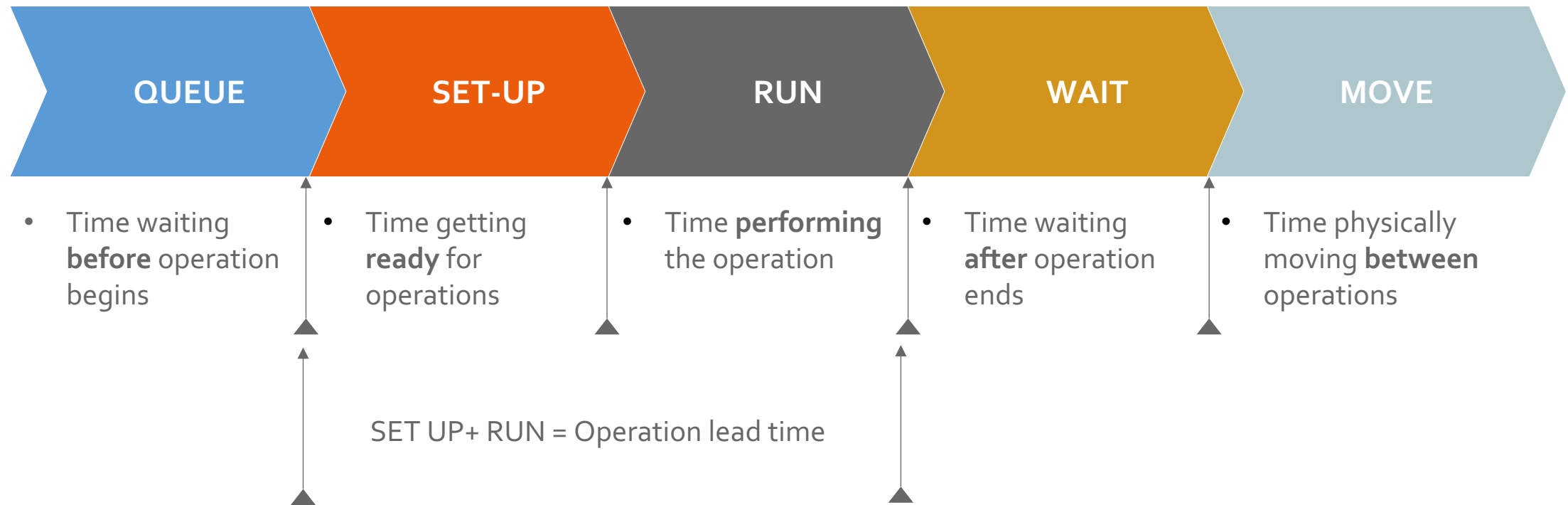
1. Calculate total load for each work center
 1. Get components requirements from MRP (qty, due dates)
 2. Determine operations time per work order (load) at each work center:

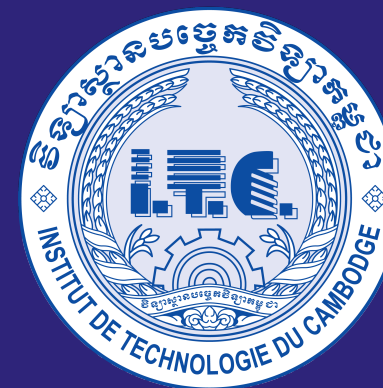
$$\text{Operation time per work order} = \text{nb of pieces} \times \text{run time per piece} + \text{set up time}$$

2. Operation time per work order are allocated to appropriate periods at each work center (process of Scheduling Operations)
3. Calculate sum of work order loads per time period at each work center

Key inputs to CRP: lead times

- MANUFACTURING lead time is composed of 5 lead times





4/ Example of calculations

Calculate the load

■ Remember:

- MRP is the key input to calculate the load.
- MRP results in a plan of production order release.

■ Example 1:

- Let's consider 4 items A,C,D,E
- All produced in the same work center
- MRP resulted in the following schedule of production order release, in pieces:

Item	Week 1	Week 2	Week 3	Week 4
A		25		50
C			100	
D			50	
E	100			

Since MRP is expressed in number of pieces, this plan must be translated into hours to **compare** it with capacity available.

Calculate the load

■ To estimate the load on a work center on a given time period, we need to :

- 1 Calculate the operation time for each **planned order and released order** at this work center
- 2 Add all the operation times together

⇒ That gives us the capacity required to meet the production schedule.

Calculate the load of a work center on week 3 | Have the vision of Planned orders + Open orders

Example 1: *We want to calculate the load of a workcenter on week 3*

We have the following data on hand, resulting from the MRP and the open shop order:

Planned order:

Item	Week 1	Week 2	Week 3	Week 4
A		25		50
C			100	
D			50	
E	100			

Open orders:

Item	Week 1	Week 2	Week 3	Week 4
A			50	
C				
D				
E				

Calculate the load of a work center on week 3 | Have the vision of Planned orders + Open orders

- There are 2 **planned orders** on week 3: 100 C and 50 D.
- We also know, based to the **open order** file, that there is 1 order for week 3: 50 A

Planned orders:

Item	Week 1	Week 2	Week 3	Week 4
A		25		50
C			100	
D			50	
E	100			

Open orders:

Item	Week 1	Week 2	Week 3	Week 4
A			50	

Warning: don't forget the open order!
⇒ In total, on week 3 you have 100 C, 50 D and 50 A

Calculate the load of a work center on week 3

| Translate quantity into hours thanks to the routing

- ➔ The quantity must be translated into hours.
- ➔ How to do it?
- ➔ **Use the information of the routing: set up time + run time**

Orders	Quantity	Setup time (h)	Run time (h/piece)	Total time (h)
A	50	0	0.5	
C	100	1	0.2	
D	50	2	0.25	
E	0	2	0.5	

⇒ The next step is to **calculate the total time for each order**

Calculate the load of a work center on week 3 | Get the total time for each order and sum everything

Orders (week 3)	Quantity	Setup time (h)	Run time (h/piece)	Total time (h)
A	50	0	0.5	25
C	100	1	0.2	21
D	50	2	0.25	14.5
E	0	2	0.5	0
Total load on this workcenter on week 3				60.5

| Compare: Load v/s rated capacity

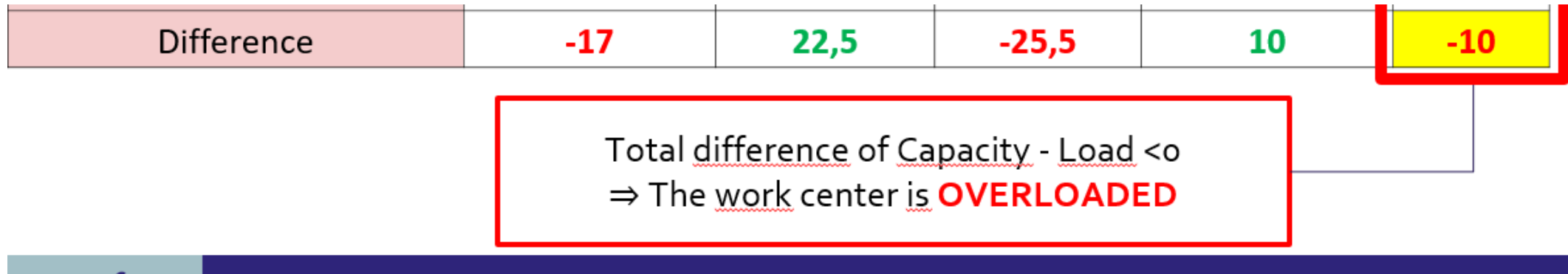
Example :let's imagine that we have calculated the load **and** rated capacity for each week (in hours: 35h/week).

➔ The load must be compared to the rated capacity in order to know whether the work center is **over loaded or under loaded within this period:**

Comparison	Week 1	Week 2	Week 3	Week 4	TOTAL
Load (open orders)			25		25
Load (planned orders)	52	12,5	35,5	25	128
Total load	52	12,5	60,5	25	153
Rated capacity	35	35	35	35	140
Difference	-17	22,5	-25,5	10	-10

Total difference of Capacity - Load < 0
⇒ The work center is **OVERLOADED**

| Propose actions to balance load & rated capacity



Conclusion: the production planner/scheduler knows that overall there is **not enough available capacity** to meet the plan (10h of capacity are missing).

➔ Different actions can be taken:

- **Rework the plan** (postpone non-priority orders)
- **Add capacities** if necessary (overtime, subcontracting...)
- **Smooth the load** on each week if possible.

Warning:
 /\ Open orders cannot be
 rescheduled /\

